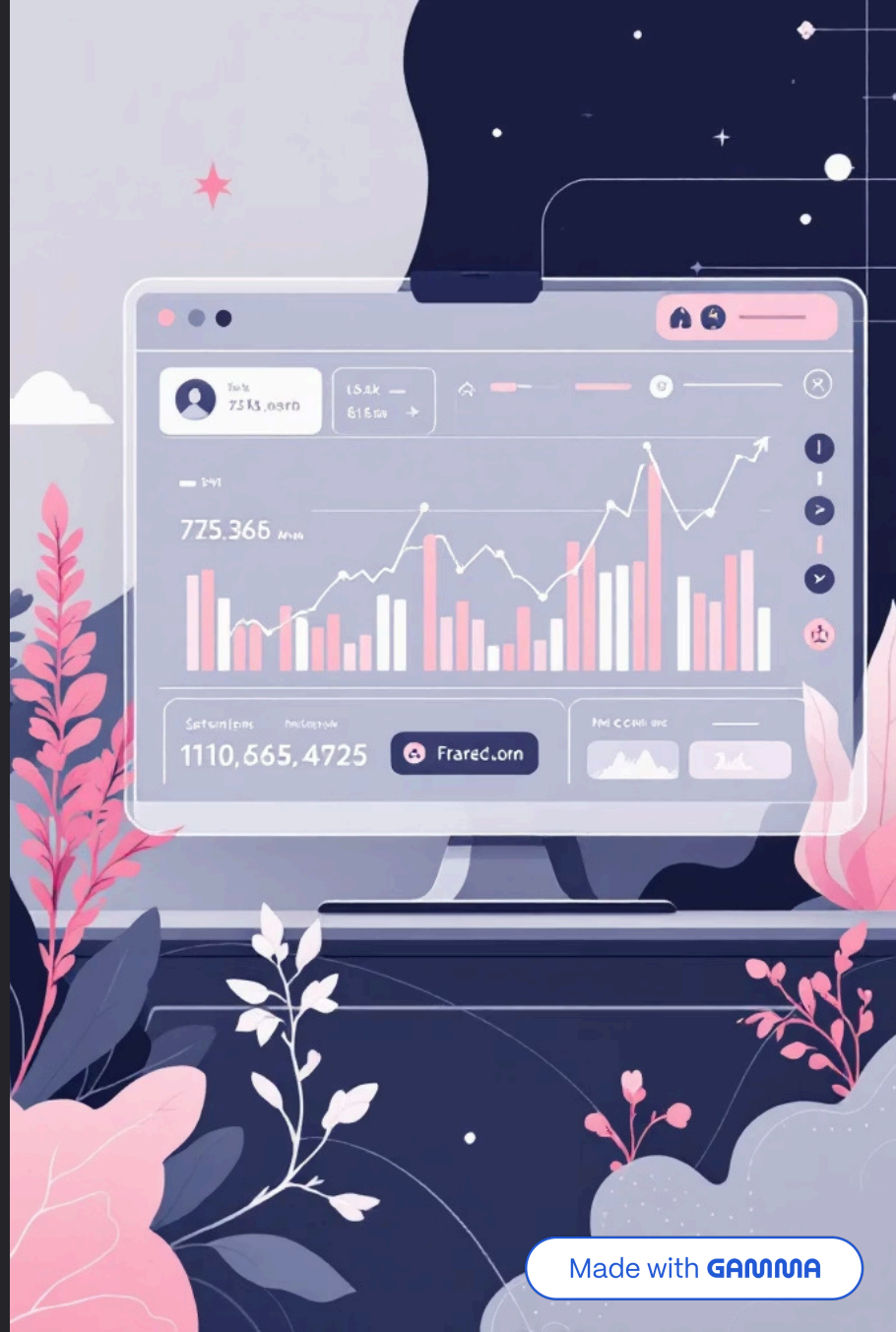
Regime-Based Trading Performance Analysis Using Market Sentiment

Uncovering the impact of market emotions on trading profitability and frequency.



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Project Objective & Data Overview

Project Objective

To analyze how trading performance varies across different market sentiment regimes (Extreme Fear → Extreme Greed) and to determine whether trading frequency impacts profitability under these emotional market conditions.

Dataset Used

- Historical trade data (Closed PnL, trade size, timestamp, buy/sell)
- Market sentiment classification: Extreme Fear, Fear, Neutral, Greed, Extreme Greed

Sentiment data was meticulously merged with trade data based on the corresponding date for precise analysis.

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Methodology: From Raw Data to Actionable Insights

Step 1: Data Cleaning

- Removed missing values
- Converted timestamps to datetime objects
- Extracted trading dates

Step 3: Trader Segmentation

Traders were categorized for comparative analysis:

- **By Frequency:** Frequent vs. Infrequent traders
- **By Profitability:** Profitable (above median PnL) vs. Less Profitable (below median PnL)

1 2 3

Step 2: Feature Engineering

Created key performance indicators:

- Daily PnL
- Win rate
- Trade frequency per account
- Profitability segmentation (based on total PnL median)

This systematic approach ensured robust data preparation and meaningful segmentation for the subsequent analysis.

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Analysis Performed: Uncovering Performance Differences

Average PnL by Sentiment & Frequency

A pivot table and heatmap visualization were created to intuitively display the average Profit and Loss across various sentiment regimes and trading frequencies, highlighting performance disparities.

Statistical Testing

Independent t-tests were conducted to compare the performance of Frequent vs. Infrequent traders within each sentiment regime. All p-values were less than 0.05, indicating statistically significant differences in performance.

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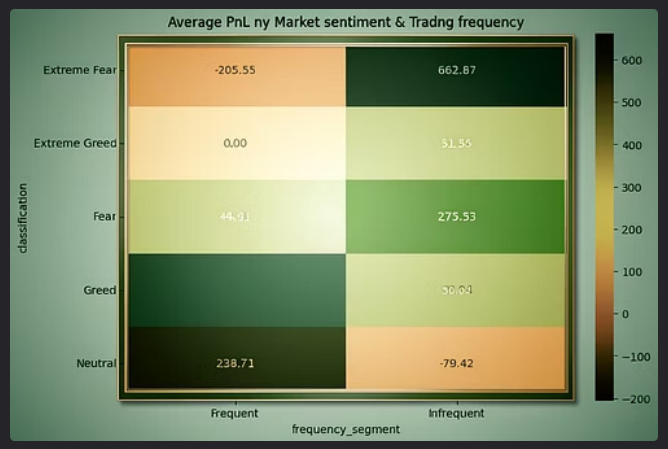

Key Findings: Performance Across Market Regimes

Extreme Fear

Frequent traders: Average Loss

Infrequent traders: Strong Profit

Result statistically significant.

- Overtrading during panic markets significantly reduces overall performance.

Fear

Infrequent traders consistently outperformed Frequent traders, with a significant difference observed.

- Selective trading is more effective in markets driven by fear.

Neutral

Frequent traders demonstrated superior performance compared to Infrequent traders.

- Active trading strategies tend to yield better results in stable, neutral market conditions.

Greed / Extreme Greed

Inconsistent performance was noted across both trader segments, suggesting heightened volatility and prevalent behavioral biases.

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Core Conclusion: Adapt to the Regime

Trading performance is demonstrably regime-dependent, with strong statistical evidence supporting that optimal trading frequency should adapt to prevailing market sentiment.

Market Regime	Suggested Strategy
Extreme Fear	Reduce trading frequency
Fear	Trade selectively
Neutral	Increase activity
Greed	Trade cautiously
Extreme Greed	Avoid overtrading

This adaptive approach can significantly enhance risk-adjusted returns.

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Strategic Insight & Business Value

Behavioral Biases Impact Outcomes

Behavioral biases profoundly affect trading outcomes, underscoring the need for discipline.

Overtrading Destroys Edge

Overtrading during emotional market conditions significantly erodes any potential trading edge.

Regime-Based Adaptation Improves Performance

Adjusting trading frequency based on market sentiment offers a proven path to improved performance.

1

Improve Risk Management

Enhance overall trading risk management strategies.

2

Build Automated Strategies

Facilitate the development of sophisticated regime-based automated trading strategies.

3

Adjust Exposure Dynamically

Enable dynamic adjustment of market exposure to optimize returns.

4

Reduce Drawdowns

Minimize drawdowns, especially during periods of market panic and heightened volatility.

This analysis strongly supports the development of sentiment-aware trading strategies for superior outcomes.

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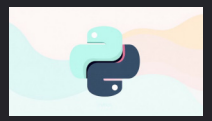
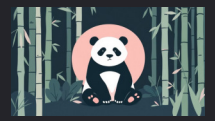
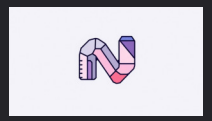
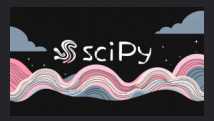
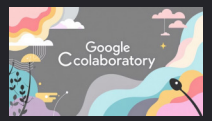
Tools & Technologies





These powerful tools facilitated the data analysis, statistical testing, and visualization required for this project.

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Explaining the Project in 60 Seconds

> “I analyzed how trading performance varies across market sentiment regimes. After merging trade data with Fear & Greed classifications, I segmented traders by frequency and profitability. I found statistically significant evidence that frequent traders underperform during fear-driven markets, while active trading works better in neutral conditions. This suggests adaptive trading frequency based on sentiment could improve risk-adjusted returns.”

A clean, professional, and data-driven summary designed for high-impact communication.

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